PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Title: Thrombophilia screening in women with recurrent first trimester miscarriage – Is it time to stop testing? – A cohort study and systematic review of the literature
AUTHORS	Shehata, Hassan; Ali, Amanda; Silva-Edge, Mariane; Haroon, Shahla; Elfituri, Abdullatif; Viswanatha, Radhika; Jan, Haider; Akolekar, Ranjit

VERSION 1 – REVIEW

REVIEWER	Kilian Vomstein Medical University Innsbruck Department of Gynecology
REVIEW RETURNED	07-Feb-2022

GENERAL COMMENTS	The study entitled “Prevalence of thrombophilia in women with recurrent early miscarriage, and a systematic review of the literature” aims to analyze the prevalence of thrombophilia in recurrent miscarriage. Furthermore, the authors perform a meta-analysis of the literature. Through the large and well-defined group of RM patients the authors provide reliable answers on the prevalence of thrombophilia in RM patients. A big caveat of the study is, that it lacks a control group. Major comments: The manuscript needs a clearer structure. As stated in the guidelines of the Journal, please stick to the main headings: Introduction, Methods, Results and Discussion Figures need legends. So far, no legends are provided by the authors. Tables are numbered in Arabic numbers, however in the text referred as latin numbers. Figures don’t have a numbering at all. The results/discussion (where does one end and the other begin?) mentions “controls”. Also, figure 1 (?) shows a control-group. However, as far as I understand the manuscript does not include a control group. If authors used a historical control group or the controls from the meta-analysis, this must be clearly stated. “Data of other inherited thrombophilia were collected but not reported due to the very rare incidence in the population.” If data was collected, it should be reported. Which other thrombophilia were recorded? Protein C/S deficiency as well as ATIII have been discussed to contribute to RM. The authors should comment on that and provide the data they collected in their large cohort. In general, I question the use of the performed meta-analysis in this case with the selections made. Minor comments: Table 3 can be deleted. The forest-plots would profit from more information: group sizes, events, weight. With that, the tables 1 and 2 could be included in the graph. Thereby, the authors could provide a table with their own data.

	Your findings suggest that there is no difference in prevalence of APLS between RM patients and controls, however you recommend the screening for APLS in RM patients in the abstract. In the discussion the authors discuss a lot concerning the definitions of an APLS, whereas it would be helpful to provide more detailed reasons on why the authors come to the clinical important conclusion in the abstract e.g. explain pregnancy complications of APLS and the easily and very cheap treatment options available.

REVIEWER	Elvira Grandone I.R.C.C.S. Casa Sollievo della Sofferenza
REVIEW RETURNED	11-Feb-2022

GENERAL COMMENTS	In this paper Shehata and coll. investigate the overall of thrombophilia in a population of women with at least 3 early losses and compare their findings to those of similar studies after a systematic review of literature. They found that 9.2% (106/1155) carried thrombophilias with most cases positive for inherited thrombophilia (8.1% , 94/1155). Only 1% (12/1155) tested positive for acquired thrombophilia. Authors conclude that, based on their study and the review of literature, testing or treatment are not recommended. The issue of pregnancy loss is relevant, as it is a relatively frequent complication with a multifactorial etiology. Few papers have considered women with the highest risk of recurrence, those with 3 or more losses. Therefore, this paper has the objective to answer a clinically relevant question. However, there are some fundamental flaws in the design of the study, that makes the results rather weak: 1) The retrospective nature can result in selection bias 2) Studies used for the systematic review are small , where only univariate analyses were performed (likely due to the small numbers). I wonder if this can sensibly influence and skew final results 3) Heterogeneity among studies was not calculated (a meta-analysis would have been more convincing)

REVIEWER	Are Pripp Oslo University Hospital, Department of Biostatistics, Epidemiology & Health Economy
REVIEW RETURNED	06-Mar-2022

GENERAL COMMENTS	Statistical review: I have focused on the statistical analysis and the presentation of results. The other reviewers and the editors need to access the medical relevance and impact. To the best of my knowledge, the statistical analysis is correct and appropriate. I have only a few minor comments. The authors might present funnel plots or use other methods to investigate publicaiton bias. They present heterogeneity statistics as Cochrane's Q and I2. These methods could be briefly described in the method section on statistical analysis. Heterogeneity statistics and the weighting of the studies could be shown in each forest plot. I was not able to find figure legends that identify and describe each figure. Please check if those are in the submitted manuscript.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1:

Dr. Kilian Vomstein, Medical University Innsbruck Department of Gynecology

Comments to the Author:

The study entitled “Prevalence of thrombophilia in women with recurrent early miscarriage, and a systematic review of the literature” aims to analyze the prevalence of thrombophilia in recurrent miscarriage. Furthermore, the authors perform a meta-analysis of the literature. Through the large and well-defined group of RM patients the authors provide reliable answers on the prevalence of thrombophilia in RM patients. A big caveat of the study is, that it lacks a control group.

Response: We thank Dr Vomstein for comments. As Dr Vomstein suggests, there is paucity of published literature regarding patients with recurrent miscarriages. Much of clinical practice worldwide is based on such limited data based on such studies. The studies which report data on recurrent miscarriages are also small in their sample size. Our study therefore, adds to the literature by providing data about prevalence of thrombophilia in a large cohort (>1,000) of pregnancies with recurrent miscarriages managed in a dedicated specialist clinic and therefore adds to the current understanding of management of recurrent miscarriage, which suffers from lack of robust data.

Major comments:

Q. The manuscript needs a clearer structure. As stated in the guidelines of the Journal, please stick to the main headings: Introduction, Methods, Results and Discussion.

Response: Thank you. We have now ensured that the above structure is followed throughout.

Q. Figures need legends. So far, no legends are provided by the authors.

Response: Thank you. The figure legends are now added.

Q. Tables are numbered in Arabic numbers, however in the text referred as latin numbers. Ok noted and corrected.

Response: Thank you. This is now corrected.

Q. Figures don’t have a numbering at all.

Response: Thank you. This is now corrected.

Q. The results/discussion (where does one end and the other begin?)

Response: Thank you. This is now corrected.

Q. mentions “controls”. Also, figure 1 (?) shows a control-group. However, as far as I understand the manuscript does not include a control group. If authors used a historical control group or the controls from the meta-analysis, this must be clearly stated.

Response: Thank you. This is now clarified and a sentence has been added in the manuscript to clarify that the comparison is with historical controls from literature.

Q. “Data of other inherited thrombophilia were collected but not reported due to the very rare incidence in the population.” If data was collected, it should be reported. Which other thrombophilia were recorded? Protein C/S deficiency as well as ATIII have been discussed to contribute to RM. The authors should comment on that and provide the data they collected in their large cohort.

Response: The data for Protein S, C and antithrombin III were not available for the entire population, hence why they were not reported. A sentence clarifying that has been added in the manuscript.

Q. In general, I question the use of the performed meta-analysis in this case with the selections made.

Response: We don’t understand the question. The methodology is standard and principles of use of meta-analysis are standard. See comments from statistical reviewer below. Please kindly clarify further.

Minor comments:

Q. Table 3 can be deleted.

Response: Agreed and deleted.

Q. The forest-plots would profit from more information: group sizes, events, weight. With that, the tables 1 and 2 could be included in the graph. Thereby, the authors could provide a table with their own data.

Response: Thank you. We have now included this in the figures 3 and 4 including information about group size, events and weight. The tables are now included as supplementary tables 1 and 2 for the interested reader as they contain information about confidence intervals, incidence rate prevalence difference with 95% confidence intervals, results about heterogeneity and bias expressed as Cochrane Q (p value), I2 statistic and Bias (p value).

Q. Your findings suggest that there is no difference in prevalence of APLS between RM patients and controls, however you recommend the screening for APLS in RM patients in the abstract. In the discussion the authors discuss a lot concerning the definitions of an APLS, whereas it would be helpful to provide more detailed reasons on why the authors come to the clinical important conclusion in the abstract e.g. explain pregnancy complications of APLS and the easily and very cheap treatment options available.

Response: We have reviewed recent literature published in this regard and have modified our conclusions and recommendations. In our original manuscript, we recommended screening as this was standard empirical practice and has been for decades so in the absence of any RCTs or similar studies, and hence it was suggested so. However, the comments from Dr Vomstein and our own review of recent literature in preparing for this revised manuscript suggests that the prevalence of

APS in the general population is not dissimilar to that in the cohort of recurrent miscarriages. We analysed this in our study and now state this in the results section. Therefore, we have now amended our conclusions and recommendations in accordance with this.

Reviewer: 2

Dr. Elvira Grandone, I.R.C.C.S. Casa Sollievo della Sofferenza

Comments to the Author:

In this paper Shehata and coll. investigate the overall of thrombophilia in a population of women with at least 3 early losses and compare their findings to those of similar studies after a systematic review of literature. They found that 9.2% (106/1155) carried thrombophilias with most cases positive for inherited thrombophilia (8.1%, 94/1155). Only 1% (12/1155) tested positive for acquired thrombophilia. Authors conclude that, based on their study and the review of literature, testing or treatment are not recommended. The issue of pregnancy loss is relevant, as it is a relatively frequent complication with a multifactorial etiology. Few papers have considered women with the highest risk of recurrence, those with 3 or more losses. Therefore, this paper has the objective to answer a clinically relevant question. However, there are some fundamental flaws in the design of the study, that makes the results rather weak:

1) The retrospective nature can result in selection bias

Response: Thank you. We agree that the study is retrospective. However, majority or almost all of literature regarding recurrent miscarriage is retrospective and with relatively small sample size. There is the added issue of lack of uniformity regarding the number and type of miscarriages in these studies. To add to this, there is lack of adherence to any internationally agreed consensus criteria in any of these studies. Even though our study is a retrospective cohort study with a potential for selection bias, our study benefits from extraction of data in a dedicated specialist recurrent miscarriage clinic with defined clinical pathways, adherence to evidence based internationally agreed criteria and importantly a large sample size of patients with recurrent miscarriages, to overcome some of the biases posed. This is potentially the largest cohort of patients with recurrent miscarriage with consecutive data on thrombophilia.

2) Studies used for the systematic review are small, where only univariate analyses were performed (likely due to the small numbers). I wonder if this can sensibly influence and skew final results

Response: Thank you. We agree with the reviewer that much of the published literature for recurrent miscarriages is based on small sample sizes. Hence, we carried out this systematic review and a meta-analysis to combine the data from these small studies to derive summary estimates, which were weighted based on sample sizes and then using a random effects model for meta-analysis, which takes into account variation not just between studies but also within studies. We also used robust measures of heterogeneity and bias, such as Cochrane's Q, I^2 statistic and Bias in our analysis to ensure that we overcome as many potential biases as one can. Our study is likely to be an improvement on what is currently published in literature and demonstrates that indiscriminate testing and treatment for acquired and inherited thrombophilia should not be undertaken as this is unlikely to improve outcomes.

3) Heterogeneity among studies was not calculated (a meta-analysis would have been more convincing)

Response: We do not understand the comment. Our manuscript and tables clearly states the meta-analysis that was carried out. We also include in our tables the analysis of heterogeneity and bias i.e. Cochrane's Q, I2 statistic and Egger Bias in each meta-analysis that was carried out. Please see comments from the statistical reviewer who states that the statistical analysis undertaken with the meta-analysis is correct and appropriate. A statement regarding this has now been added in the statistical analysis in the methods section.

Reviewer: 3

Dr. Are Pripp, Oslo University Hospital

Comments to the Author: / Statistical review:

I have focused on the statistical analysis and the presentation of results. The other reviewers and the editors need to access the medical relevance and impact. To the best of my knowledge, the statistical analysis is correct and appropriate. I have only a few minor comments. The authors might present funnel plots or use other methods to investigate publication bias. They present heterogeneity statistics as Cochrane's Q and I2. These methods could be briefly described in the method section on statistical analysis. Heterogeneity statistics and the weighting of the studies could be shown in each forest plot. I was not able to find figure legends that identify and describe each figure. Please check if those are in the submitted manuscript.

Response: Thank you for your comments. Figure legends are now included to identify figures in the submitted manuscript. We now describe these in the methods section on statistical analysis. We have now included more information in the figures by including group size, event rates and weights of studies. The tables have been made supplementary as they contain information about prevalence with confidence intervals, incidence rate meta-analysis results for the interested reader.

VERSION 2 – REVIEW

REVIEWER	Elvira Grandone I.R.C.C.S. Casa Sollievo della Sofferenza
REVIEW RETURNED	24-Apr-2022

GENERAL COMMENTS	The fundamental bias of this study, as well as that of most available studies in the field, is that abnormal fetal karyotype, that is by far the most frequent cause of recurrent pregnancy loss (RPL), is completely ignored (Gu C, Li K, Li R, et al. Chromosomal Aneuploidy Associated With Clinical Characteristics of Pregnancy Loss. Front Genet. 2021;12:667697. Published 2021 Apr 15; Nikitina TV, et al. Karyotype evaluation of repeated abortions in primary and secondary recurrent pregnancy loss. J Assist Reprod Genet. 2020;37(3):517-525. doi:10.1007/s10815-020-01703;). It is also known that aneuploid rates are significantly higher in women >35 years with pregnancy loss <11 gestational weeks (Gu C, Li K, Li R, et al. Chromosomal Aneuploidy Associated With Clinical Characteristics of Pregnancy Loss. Front Genet. 2021;12:667697.

	Published 2021 Apr 15. doi:10.3389/fgene.2021.667697); authors do not take into account at all this relevant information: no analysis regarding these potential confounders is shown. Last, but not least, the only severe thrombophilia for which authors show data is the acquired one (i.e. antiphospholipid antibodies), while the severe inherited thrombophilias were not considered. It's time to design different studies and move forward in increasing the knowledge of this common and stressing event. RPL is a multifactorial disease, often being the result of gene–gene and gene-environment interactions. Therefore, the issue of foetal aneuploidy is relevant even in our attempts to answer to the question whether thrombophilia (especially the most common genetic variants, as Factor V Leiden and FIIG2020A) is truly associated with a higher risk of RPL. It is widely known that polymorphic variants (i.e.: the inheritance of a trait controlled by a single genetic locus with two alleles, in which the least common allele has a frequency of about 1% or greater) with low prevalence in the population have a relatively low impact on the absolute risk of disease. "A simple model expressing the presence or the absence of two dichotomous factors (high risk allele and exposure to an environmental risk factor) is often not sufficient due to incomplete clinical penetrance and variable expressed severity.As such, it is not surprising that carriers of FV Leiden or FIIG20210A can have uneventful pregnancies more often than obstetric complications " (see Grandone E, Piazza G. Reply: The pathway to the 'truth' in the study of recurrent pregnancy loss and thrombophilia. Hum Reprod. 2021 Dec 27;37(1):191-193).These fundamental biases in the study here shown will continue to distort the reality; the effort we should make to advance our knowledge and help our patients is to start to design adequate studies, avoiding to use those with so relevant biases.

REVIEWER	Are Pripp Oslo University Hospital, Department of Biostatistics, Epidemiology & Health Economy
REVIEW RETURNED	13-Apr-2022

GENERAL COMMENTS	Thank you for a thorough revision of the manuscript. My previous comments about the statistical analysis have been addressed and I recommend the manuscript for publication.

VERSION 2 – AUTHOR RESPONSE

Reviewer 1: There are no comments received.

Comments to the Author: None received.

Our response: No comments received.

Reviewer: 3, Dr Are Pripp, Oslo University Hospital

Comments to the Author: Thank you for a thorough revision of the manuscript. My previous comments about the statistical analysis have been addressed and I recommend the manuscript for publication.

Our response: Thank you for appreciating that we have resubmitted a thoroughly revised version of the manuscript which is statistically sound and addresses all concerns.

Reviewer 2: Dr. Elvira Grandone, I.R.C.C.S. Casa Sollievo della Sofferenza

The fundamental bias of this study, as well as that of most available studies in the field, is that abnormal fetal karyotype, that is by far the most frequent cause of recurrent pregnancy loss (RPL), is completely ignored (Gu C, Li K, Li R, et al. Chromosomal Aneuploidy Associated With Clinical Characteristics of Pregnancy Loss. Front Genet. 2021;12:667697. Published 2021 Apr 15; Nikitina TV, et al. Karyotype evaluation of repeated abortions in primary and secondary recurrent pregnancy loss. J Assist Reprod Genet. 2020;37(3):517-525. doi:10.1007/s10815-020-01703;). It is also known that aneuploid rates are significantly higher in women >35 years with pregnancy loss <11 gestational weeks (Gu C, Li K, Li R, et al. Chromosomal Aneuploidy Associated with Clinical Characteristics of Pregnancy Loss. Front Genet. 2021;12:667697. Published 2021 Apr 15. doi:10.3389/fgene.2021.667697); authors do not take into account at all this relevant information: no analysis regarding these potential confounders is shown. Last, but not least, the only severe thrombophilia for which authors show data is the acquired one (i.e., antiphospholipid antibodies), while the severe inherited thrombophilias were not considered.

It's time to design different studies and move forward in increasing the knowledge of this common and stressing event. RPL is a multifactorial disease, often being the result of gene–gene and gene-environment interactions. Therefore, the issue of foetal aneuploidy is relevant even in our attempts to answer to the question whether thrombophilia (especially the most common genetic variants, as Factor V Leiden and FIIG2020A) is truly associated with a higher risk of RPL. It is widely known that polymorphic variants (i.e.: the inheritance of a trait controlled by a single genetic locus with two alleles, in which the least common allele has a frequency of about 1% or greater) with low prevalence in the population have a relatively low impact on the absolute risk of disease. "A simple model expressing the presence or the absence of two dichotomous factors (high risk allele and exposure to an environmental risk factor) is often not sufficient due to incomplete clinical penetrance and variable expressed severity. As such, it is not surprising that carriers of FV Leiden or FIIG20210A can have uneventful pregnancies more often than obstetric complications " (see Grandone E, Piazza G. Reply: The pathway to the 'truth' in the study of recurrent pregnancy loss and thrombophilia. Hum Reprod. 2021 Dec 27;37(1):191-193). These fundamental biases in the study here shown will continue to distort the reality; the effort we should make to advance our knowledge and help our patients is to start to design adequate studies, avoiding using those with so relevant biases.

Our response: We are concerned, and a bit confused about this response. We do not believe that the peer reviewer has appreciated the context of our manuscript. The original peer review of our manuscript raised some valid points which we addressed in our revised version. There was correctly no reference to a genetic aetiology in these cohort of patients. But then in complete contrast to the original peer review, the subsequent review of our revised version seems to have ignored all the aspects which we addressed in our revised version and has now raised the association with genetic abnormalities as an etiological factor in recurrent miscarriages.

We not only believe this to be scientifically inaccurate; but also, the peer reviewer seems to have completely missed the point of our manuscript. Our study does not address the issue of aetiological factors in recurrent miscarriage, but specifically looks at inherited and acquired thrombophilia as a cause and raises the fact that the prevalence of this particular factor is not dissimilar to the general population.

Our study proposes and provides new scientific evidence based on a large cohort of recurrent miscarriage couples and challenges current clinical practice, which is based on small studies. In addition to our cohort, we also undertake a systematic review of literature and a meta-analysis based on robust statistical principles, as outlined by your own statistical peer-reviewer.

In addition, the peer reviewer provides unrelated and inappropriate references to support this poorly conducted peer review. For example, they quote a paper by Gu et al., 2021 (Chromosomal Aneuploidy Associated With Clinical Characteristics of Pregnancy Loss Front Genet. 2021;12:667697). This particular study addresses the issue of aneuploidies and sporadic pregnancy losses and not recurrent miscarriage. Additionally, the study only addressed the underlying causes of sporadic pregnancy loss, and only reports results for those which underwent genetic analysis.

It must be noted that not all pregnancies that undergo pregnancy losses are sent for genetic analysis, so the prevalence of genetic causes is overestimated in studies that only examine a select cohort who underwent genetic testing, and this does not address population prevalence in all miscarriages. More importantly, we are concerned that a reason for rejection of our study was based on a reference to a biased study reporting the association of genetic defects with 'sporadic' and not recurrent pregnancy losses is very misleading, and highlights that the peer reviewer has not appreciated the objective, the study design, our methodology and the conclusions.

Similarly, the second paper quoted by the peer reviewer (Nikitina TV, et al. Karyotype evaluation of repeated abortions in primary and secondary recurrent pregnancy loss. J Assist Reprod Genet. 2020;37(3):517-525) is an example of why our study was conducted. This study was based on small number of 108 cases of recurrent embryo/fetal loss with a vague definition of pregnancy loss, which is not consistent with the internationally accepted definition of recurrent miscarriages. The medical literature is fraught with anecdotal data from small studies which do not follow internationally agreed definitions. Therefore, it was necessary for a study such as ours with a large cohort managed in a specialist centre, along with a systematic review of literature of studies which followed international guidelines.

The peer review and the comments made are in direct contrast to the recommendations from international professional bodies. The European Society of Human Reproduction and Embryology (ESHRE) in their recent guidance on recurrent pregnancy loss explicitly state that genetic testing in couples with recurrent miscarriages should only be carried out after an individual risk assessment as abnormal parental karyotypes were only found in less than 2% of individuals. Routine and indiscriminate genetic testing is unlikely to be beneficial as the chances of finding an abnormality are very low.

Given this background, we are quite surprised to read the reviewer comments stating that genetic and chromosomal abnormalities are by far the most common cause of recurrent pregnancy loss. As highlighted above, this certainly is not the case based on well-conducted large cohort studies and is also reflected in national and international guidance. Moreover, it is certainly not the objective of our study, which was specifically examining the prevalence of inherited and acquired thrombophilia in a large cohort of couples with recurrent miscarriage and we combined that with a systematic review of literature and meta-analysis based on standard statistical principles. The data presented in our study is from a specific dedicated recurrent miscarriage clinic where the care is led and delivered by experts in the field. We are concerned that the peer reviewer has not appreciated our study objectives, study

design and methodology from a scientific point of view but merely alluded to lack of inclusion of genetic factors as a basis for rejection, which is of concern.

We are also of the opinion that this poor peer review should have been identified in the editorial review process.

The BMJ is a widely read and prestigious peer-reviewed scientific journal, and we hope that our feedback is reviewed by the Editors and the inappropriate rejection of an important study is reconsidered.